

AI-Based Remedial Class Scheduler – Product Requirements Document

■ Objective

Build a web application that automatically schedules remedial classes for students scoring below 10 marks, by matching teacher and student availability using GenAI reasoning.

■ Core Features

- Data Management: CRUD operations for students, teachers, and timetables.
- AI Scheduler: Uses LLM to generate conflict-free remedial timetables.
- Frontend Dashboard (React + Tailwind): Manage and display data visually.
- AI Integration: LangChain + GPT/Gemini for intelligent scheduling.
- Validation: Avoids time conflicts and ensures fairness.

■■ Tech Stack

Layer	Technology
Frontend	React.js + Tailwind CSS
Backend	FastAPI (Python)
Database	MySQL (via SQLAlchemy ORM)
AI	LangChain + OpenAI/Gemini API
Tools	Axios, Pandas, ReportLab, Chart.js

■ System Architecture

React + Tailwind (Frontend) → FastAPI Backend → LangChain + LLM → MySQL Database

■■ Database Schema

Tables: Students, Teachers, Timetable

■ AI Logic Flow

- Filter students with marks < 10.
- Match free slots between teachers and students.
- Send structured data to LLM for schedule generation.
- Validate and save generated timetable.
- Display results on the frontend.

■ Acceptance Criteria

- Students with marks < 10 identified automatically.

- Generated timetable is conflict-free.
- Timetable saved and displayed correctly.
- APIs and frontend connected successfully.
- Runs locally using MySQL with minimal setup.